Problem

The sum of 23 and double a different number is 55. Form and solve a linear equation to find the value of the unknown number.

Solution

Let the unknown number be x.

$$23 + 2x = 55$$

$$-23 \qquad 2x = 32 \qquad -23$$

$$\div 2 \qquad x = 16 \qquad \div 2$$

The unknown number is 16.

Remember, you can check your answer by substituting it into the original question and checking that it works.

Problem

Ben's sister is 4 years older than him. The sum of the two children's ages is 22 years. Form and solve a linear equation to work out Ben's age.

Solution

Let Ben's age be x.

$$x + x + 4 = 22$$

$$2x + 4 = 22$$

$$-4 \qquad 2x = 18 \qquad -4$$

$$\div 2 \qquad x = 9 \qquad \div 2$$

Ben is 9 years old

Remember, you can check your answer by substituting it into the original question and checking that it works.

Problem

The length of a rectangle is twice the value of its width. The perimeter of this rectangle is 36cm. Form and solve a linear equation to find the dimensions of the rectangle.

Solution

Let the width of the rectangle be equal to $x$cm. Then, its length is $2x$cm.

$$\text{Perimeter} = x + 2x + x + 2x = 6x$$

$$6x = 36$$

$$\div 6 \qquad x = 6\text{cm} \qquad \div 6$$

The width is 6cm, so the length is 12cm.

Remember, you can check your answer by substituting it into the original question and checking that it works.

Problem

Charlie has some marbles. Bella has 4 fewer marbles than Charlie. Edwin has 9 more marbles than Charlie. The children have 83 marbles altogether. Work out how many marbles Charlie has.

Solution

Let the number of marbles Charlie has equal x. Bella has $x - 4$ marbles and Edwin has $x + 9$ marbles.

$$x + x - 4 + x + 9 = 83$$

$$3x + 5 = 83$$

$$-5 \qquad 3x = 78 \qquad -5$$

$$\div 3 \qquad x = 26 \qquad \div 3$$

Charlie has 26 marbles.

Remember, you can check your answer by substituting it into the original question and checking that it works.